

Introduction to Computing (CS 101)

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Outline

- CS101 - Introduction to Computing [L-T-P-C: 3-0-0-6]
- Instructors for Pre-Mid Sem Part
- John Jose (1, 3) & T. Venkatesh (2, 4)
- Two exams: Mid semester (50%), End semester(50%)
- Biometric attendance policy – 75% mandatory
- Late arrival in class not allowed
- CS101 Teams Group for slides upload and discussions
- Medical Leave Records to upload (Teams) within 3 days

Instructors and TAs

Division	Branches
I	ME, CE, ENERGY
II	BT, ECE, EEE
III	CSE, MC, EPH
IV	CL, CST, DSAI, DD

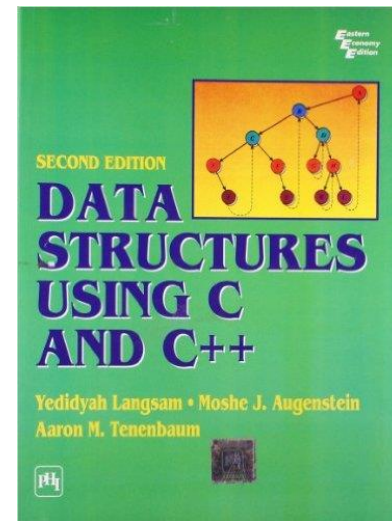
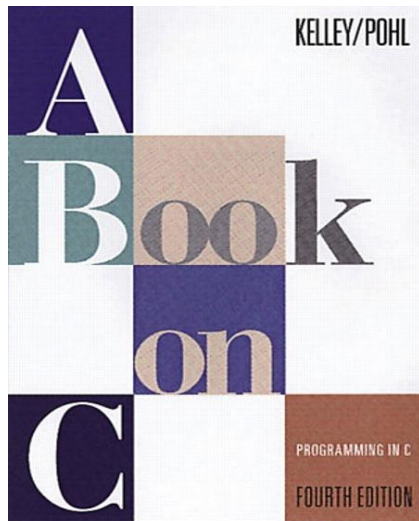
Instructor	Section / Slot	Hall	Lead TA	Associate TAs [Alternate Weeks]	
JJ	Section 1 (FN) -D Slot	L2	Vivekananda G	Soumya Bharadwaj	Prince Yadav
TV	Section 2 (FN)- D Slot	L4	Abir Banerjee	Nilotpola Sarma	Debarshi Roy
JJ	Section 3 (AN) -D1 Slot	L2	Soumya Asati	Imlijungla Longchar	K Chitra
TV	Section 4 (AN)- D1 Slot	L4	Subramanian V	Neeraj Sharma	Swati Upadhyay
Timing - Sections: 1 and 2 [D]			Timing - Sections: 3 and 4 [D1]		
Mon : 11 00 to 11:55 AM			Mon : 2:00 to 2:55 PM		
Thu: 9:00 to 9:55 AM			Thu: 4:00 to 4:55 PM		
Fri: 10:00 to 10:55 AM			Fri: 3:00 to 3:55 PM		

Syllabus (Pre Mid-Sem Part)

- Introduction: Von Neumann architecture, machine language, assembly language, high level programming languages
- Compiler, interpreter, loader, linker, text editors, operating systems
- Flowchart; Basic features of programming in C, data types, variables, operators, expressions, statements, control structures, functions.

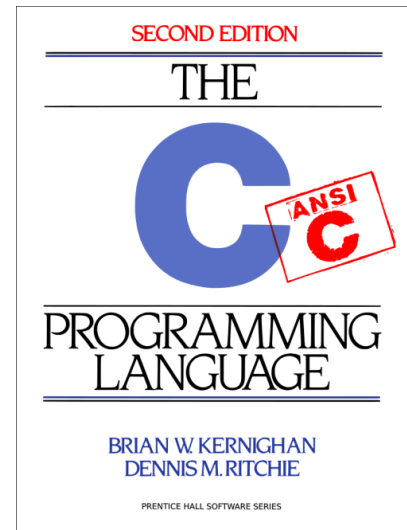
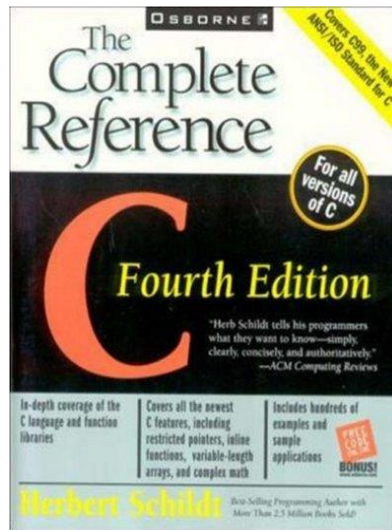
Text Book

- A. Kelly and I. Pohl, *A Book on C*, 4/e, Pearson Education, 1999
- A M Tenenbaum et al, *Data Structures Using C*, Prentice Hall India, 1996.



References

- H Schildt, **C: The Complete Reference**, 4/e, Tata Mcgraw Hill, 2000
- B Kernighan and D Ritchie, **The C Programming Language**, 2/e, Prentice Hall of India, 1988.



- You many refer/read any book/source you like

CS101 : Course Flow

Computer System Fundamentals

Basic C Programming

Advanced C Programming

Data Structures &
Sorting /Searching

Understanding the Computer



- A computer is a machine that, under a program's direction, performs four basic operations: input, processing, output, and storage.
- A program is a list of instructions that tells a computer how to perform the four operations.

What Is A Computer?

- An electronic device
- Operates under control of instructions (software) stored in its own memory unit
- It can
 - Accept data (input),
 - Manipulate data (process),
 - Produce output from the processing.
- A collection of devices that function together as a system.

Input



- The first operation.
- The computer inputs data from an input device.
- Electrical impulses representing words, numbers, images, or sounds.

Processing

Arithmetic

$$4 + 16 = 20$$

Logic

$$728 > 546$$

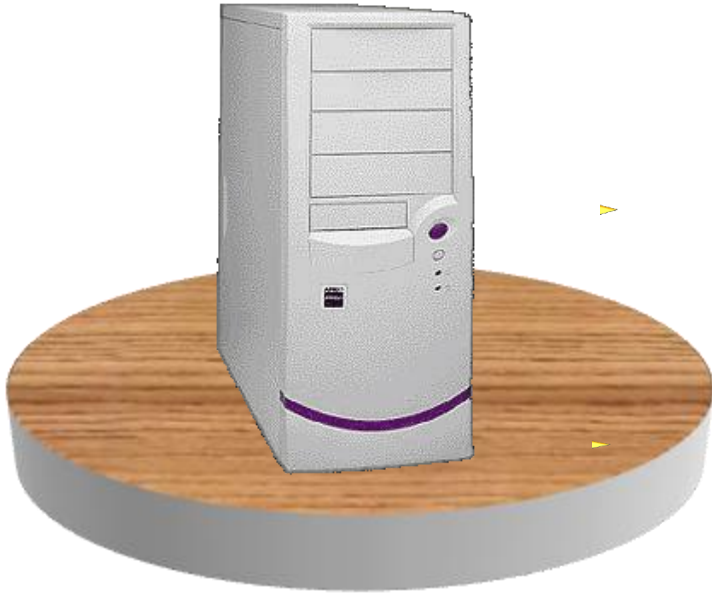
- The second operation.
- The computer performs arithmetic or comparison (logical) operations on the data.
- Performed at a very high speed.

Output



- The third operation.
- The computer shows the results of the processing operation in a way people can understand.
- Data is processed into information.

Storage



- The fourth operation.
- The computer saves the data or output so that it can be used again later.

Computer System



Hardware



Software

- A collection of related computer components that are designed to work together.
- A system includes hardware and software.

The Information Processing Cycle

Input **Processing** **Output** **Storage**

- The computer performs the four basic operations (input, processing, output, and storage).
- Two types of processing cycles:
 - **Batch processing**— A lockstep sequence used by older computers.
 - **Interactive processing**— Initiating several information processing cycles in a single session; used by modern computers.

Hardware

- Hardware is the computer's physical components.
 - **Input devices**— Enable users to enter data into the computer for processing.
 - **Processing circuitry**— Components located in the system's case that transform data into information.
 - **Output devices**— Peripherals that show the results of processing.
 - **Storage devices**— Used to store all the programs and data that the computer uses.
 - **Communication Devices**— Used to move data between computers.

Input: Key Terms

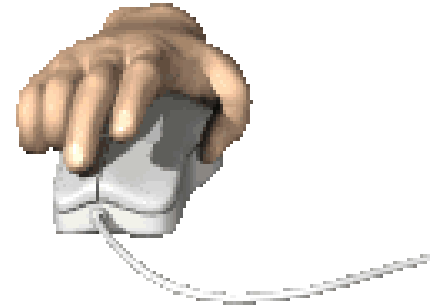
- **Command**— A type of input that tells the program what to do.
- **Confirmation**— A message that indicates if the command was or was not carried out.
- **Keyboard**— Enables the user to input characters.
- **Pointing device**— Enables the user to move an on-screen pointer.
- **Speech-recognition**— A program that enables the computer to transcribe spoken words into text.

Input Devices

Keyboard



Mouse



Microphone



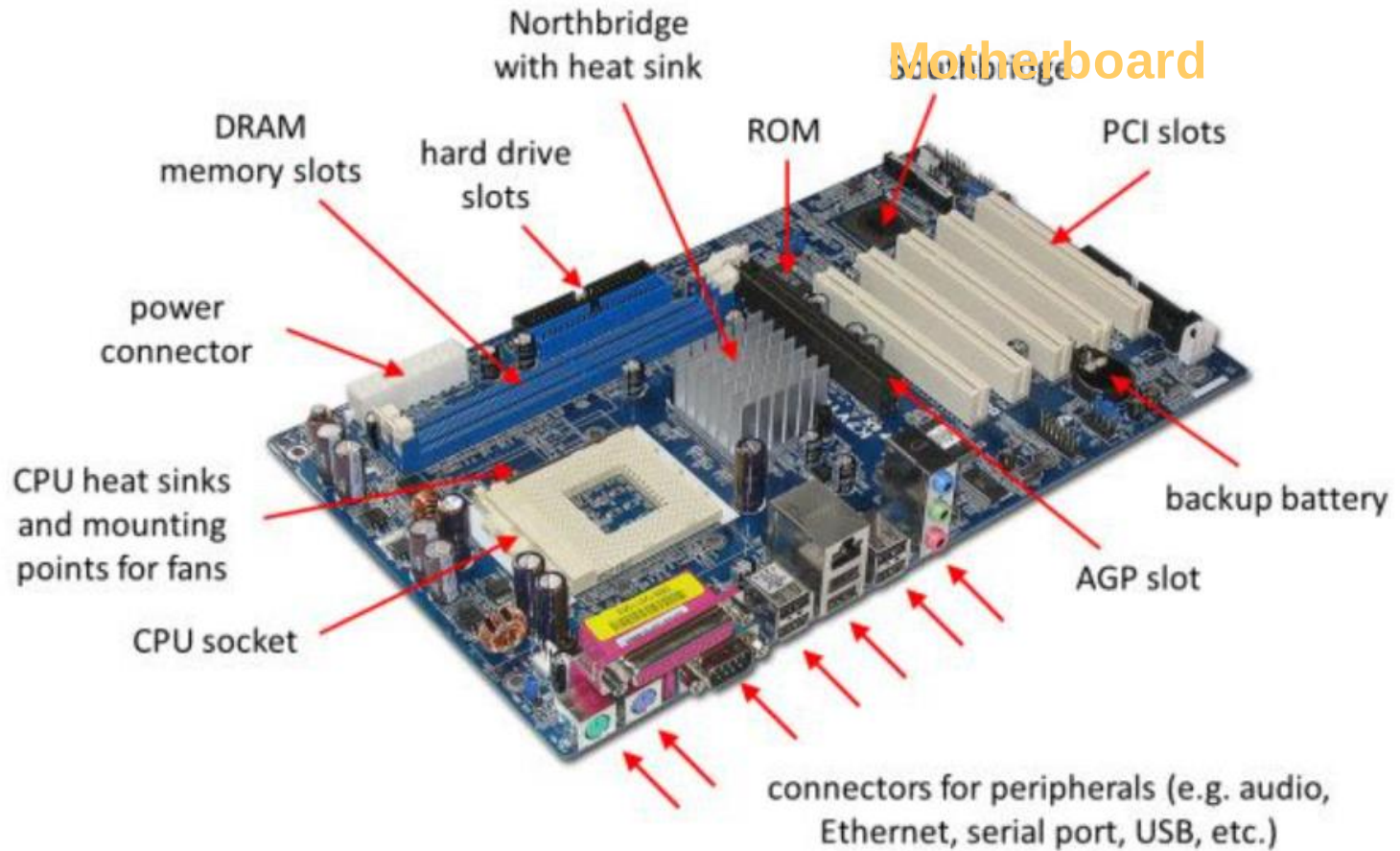
Digital Cameras



Processing: Key Terms

- **Central processing unit (CPU)**– The computer's processing circuitry. It is also called a microprocessor.
- **Motherboard**– The computer's main circuit board.
- **Memory**– Chips that store program instructions. Random access memory (RAM) is the most important type.
- **Expansion slots**– A receptacle designed to accept a plug-in expansion card.
- **Expansion card**– Used to connect peripherals (input/output devices) to the motherboard.

Processing Devices



RAM



Microprocessor



Output: Key Terms

- **Monitors**— Display processing results on screen.
 - Cathode ray tube— Electron Gun screens
 - Flat-panel displays (LCD)—liquid crystal diodes
- **Speakers**—Results as sound.
- **Printers**— Generate output on paper.
 - **Impact printers**— Print heads strike a ribbon to produce an image.
 - **Nonimpact printers**— Transfer the image to the paper.
Inkjet printers, Laser printers

Output Devices

Monitor



Printers



Speakers

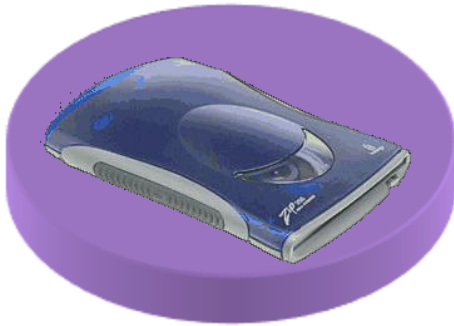


Storage: Key Terms

- **Magnetic storage media**– Uses magnetically sensitive material.
- **Optical storage media**– Uses a laser beam to scan pits etched into a disc.
- **Solid State Media Drives** (pen drive) – uses properties of materials for permanent storage.

Storage Devices

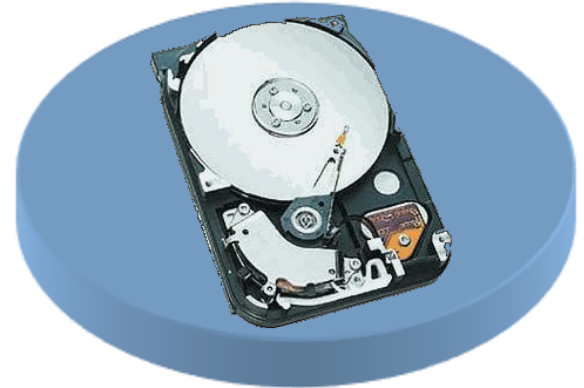
Zip Drive



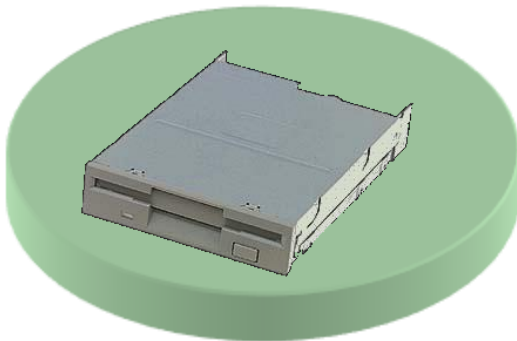
CD / DVD Drive



Hard Drive



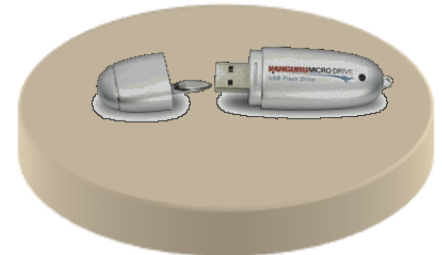
Floppy Disk Drive



Tape Drive



Pen Drive



Communications Devices: Key Terms

- **Computer network:** Two or more computers linked by some type of connection.
- **Network interface card (NIC):** Enables computers to connect to each other in a limited geographic area.
- **Local area network (LAN):** Computers connected to each other within a building or group of buildings.

Types of Computers

Desktop



Workstation



Laptop



PDA, Mobile phone



Introduction to Software

- Software consists of collection of programs.
- Program consists of sequence of instructions.
- Programs is stored as a files.
- Software is created by programmers using a programming language.
- Transferring a program into the computer's memory is called loading.
- Once loaded into memory the program's instructions are carried out or executed.

Programs

High Level Language Program

```
#include<stdio.h>
int main()
{
    double a, b , result =0 ;
    printf("Enter first number : ");
    scanf("%lf",&a);

    printf("Enter Second Number : ");
    scanf("%lf",&b);

    result = a + b;
    printf("Addition of two numbers is : %lf", result);

    return 0;
}
```

Machine Code

```
0011000000000000 ; read n -> acc ;
1011000000001010 ; jump to Done if n < 0. ;
01010000000010000 ; add sum to the acc ;
00100000000010000 ; store the new sum ;
10010000000000000 ; go back & read in next number ;
00010000000010000 ; load the final sum ;
01000000000000000 ; output the final sum ;
00000000000000000 ; stop ;
00000000000000000 ; 2-byte location where sum is stored ;
00000000000000000 ;
00000000000000000 ;
00000000000000000 .
```

Assembly Code

```
CODESEG SEGMENT 'CODE'
    ASSUME CS:CODESEG
ZERO    PROC FAR
    PUSH DS
    MOV AX,0H
    PUSH AX
    MOV AX,2000H
    MOV DS,AX
    MOV DI,0H
    MOV CX,00FFH

START:  MOV DX,0H
        MOV [DI],DX
        INC DI
        DEC CX
        JNZ START
        RET
ZERO    ENDP
CODESEG ENDS
END
```

Types of Software

System software



Application software



- **System software**— All programs that help the computer function properly.
- Provides support for running application software.
 - Operating system, Loader, Linker, Compiler
- **Application software**— All the utility programs you use to perform a task.
 - Text editor, browser, calculator, spreadsheet

Types of Operating Systems

- **Command-line interface**– Users type instructions at the keyboard, one line at a time. Examples: MS-DOS, UNIX-shell
- **Graphical user interface (GUI)**– Users choose items from menus by using a pointing device to click on icons which represent resources and commands. Examples: Windows, Mac OS

Command-Line Interface

```
C:\ Command Prompt
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

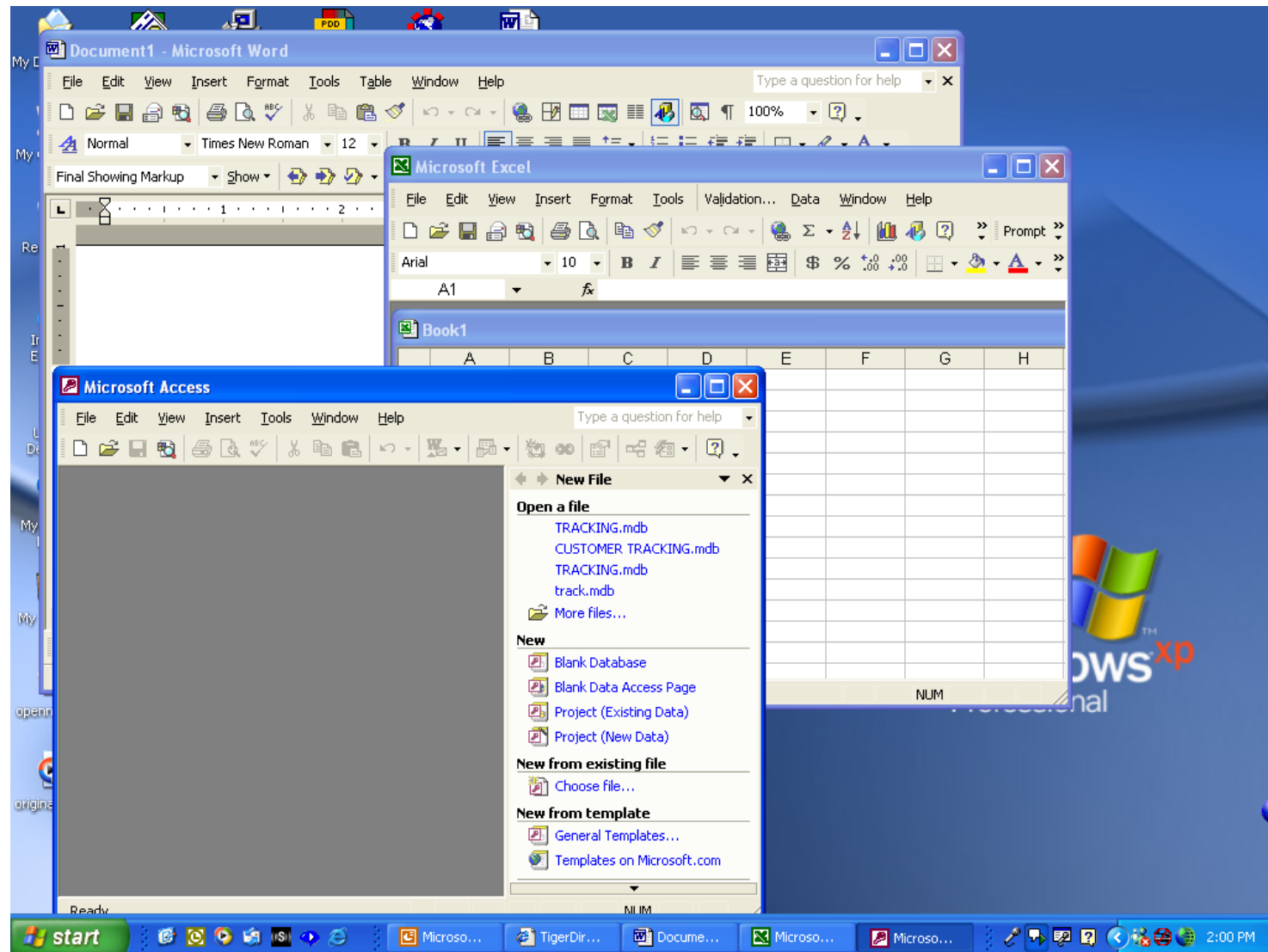
C:\Documents and Settings\default>dir/p
Volume in drive C has no label.
Volume Serial Number is 07D0-0A0D

Directory of C:\Documents and Settings\default

02/12/2002  09:53 AM    <DIR>          -
02/12/2002  09:53 AM    <DIR>          ..
02/12/2002  10:21 AM    <DIR>          My Documents
02/12/2002  10:21 AM    <DIR>          Favorites
02/12/2002  09:35 AM    <DIR>          Desktop
02/12/2002  09:35 AM    <DIR>          Start Menu
02/14/2002  09:45 PM    <DIR>          WINDOWS
               0 File(s)                0 bytes
               7 Dir(s)  11,026,939,904 bytes free

C:\Documents and Settings\default>_
```

Graphical User Interface (GUI)



Application Software

- Application software includes all programs that enable us to use the computer in a useful way.
- Custom software is developed for specific needs.
- Packaged software is produced for the mass market.

Software Suite



MS WORD

MS EXCEL

MS POWERPOINT

MS ACCESS

MS FRONT PAGE

- A collection of full-featured standalone programs that usually share a common command structure and have similar interfaces.

Thanks